



Designing for People Who Are Blind

Karin (54 years old) is married and has been blind since birth. She has a guide dog and a cane. She has three teenagers and lives in Amsterdam. Karin has a steady job in The Hague, which she commutes to independently. Unexpected obstacles on her familiar routes, such as shared e-scooters left on the sidewalk, make her journey more difficult. When there are disruptions or changes in public transport, travel apps assist her. At the office, she works efficiently with her braille display and screen reader software, even though some programs are not fully accessible.



Use

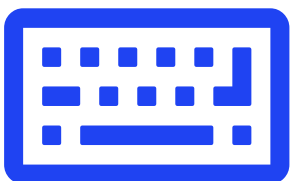
Design that works with voice navigation or screen readers



Clear heading structure using HTML

`<h1>`
`<h2>`
`<h3>`

Keyboard-accessible navigation



Descriptive links and buttons



Alternative text for images and audio descriptions for videos

`<alt>`



Avoid

Conveying information only through video or images



Structuring documents only through text formatting

Header
Paragraph

Designs that require mouse only use

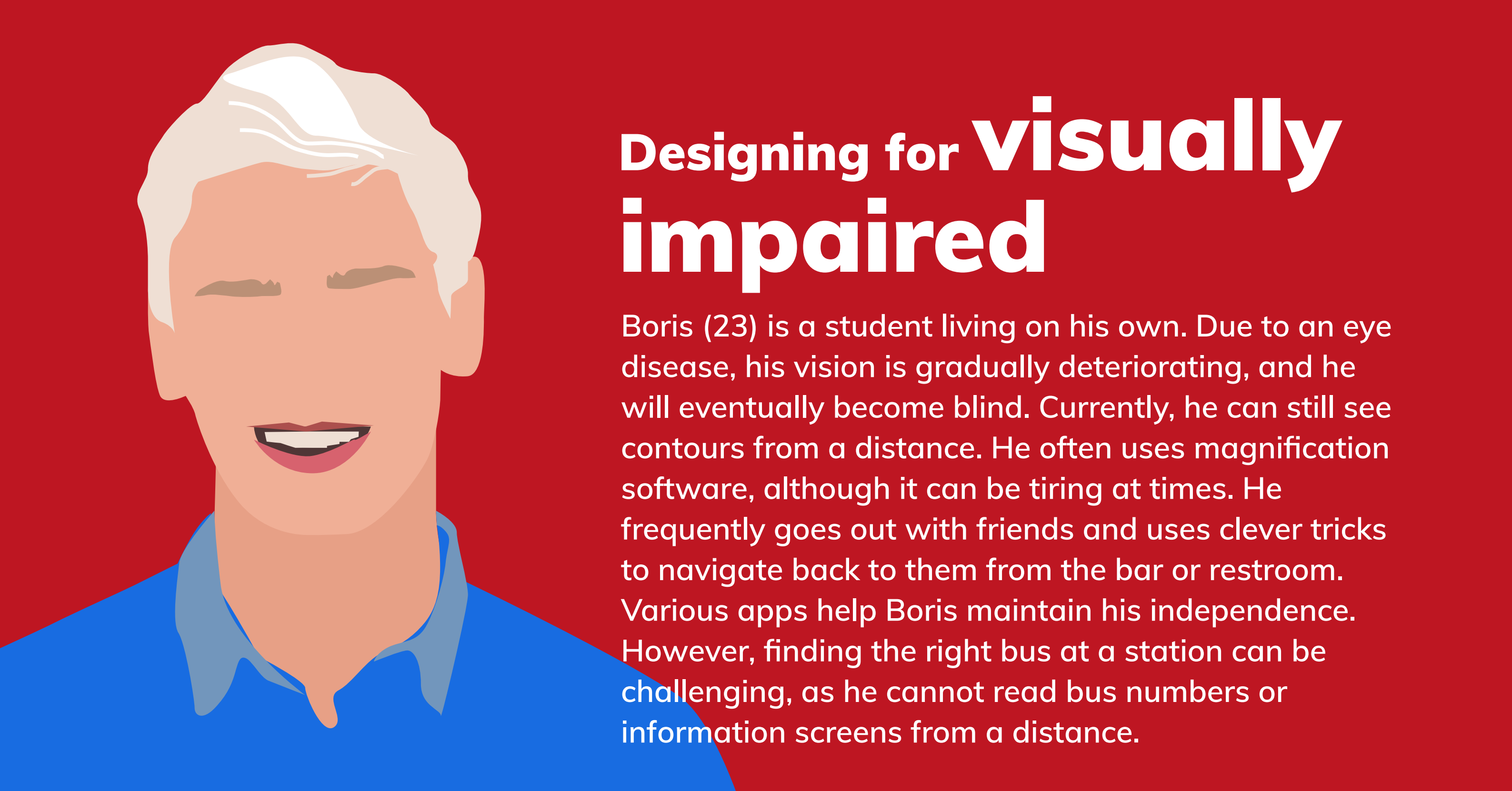


Links and headings without information



Notifications that briefly appear on screen and are not read aloud by screen readers





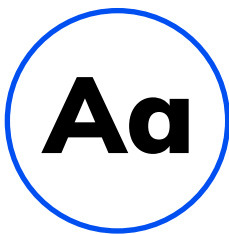
Designing for visually impaired

Boris (23) is a student living on his own. Due to an eye disease, his vision is gradually deteriorating, and he will eventually become blind. Currently, he can still see contours from a distance. He often uses magnification software, although it can be tiring at times. He frequently goes out with friends and uses clever tricks to navigate back to them from the bar or restroom. Various apps help Boris maintain his independence. However, finding the right bus at a station can be challenging, as he cannot read bus numbers or information screens from a distance.



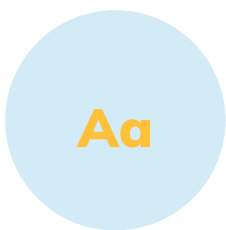
Use

Clear color contrast and a readable font-size



Avoid

Low color contrast and small font sizes



HTML to publish all information



Hiding information in downloads



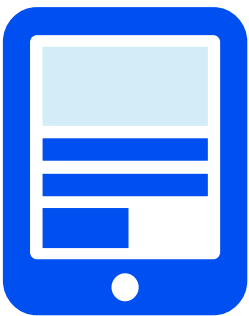
A combination of color, shape, and text



Relying only on color to convey meaning



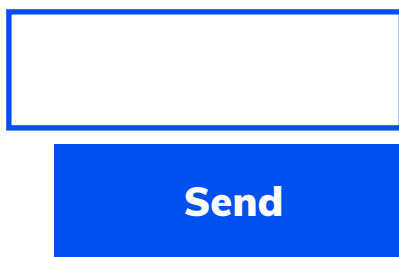
A linear, logical layout



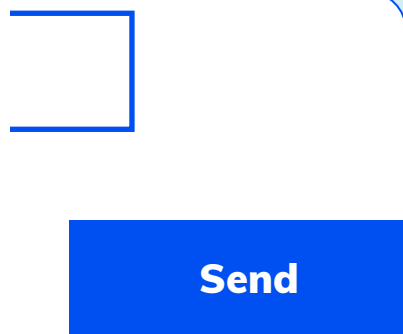
Spreading content across the entire page



Buttons and notifications in context



Space between actions and context





Designing for People Who Are Color-Blind

Samir (13) lives in a small village in Zeeland. He is in his first year of secondary school and plays football three times a week. His favorite apps are TikTok, Snapchat, and the KNVB app. Samir is color-blind. This mainly causes difficulties in subjects like geography and mathematics, where he needs to read graphs and tables. He enjoys gaming, but not all games take color blindness into account. Allies are often green and opponents red, so he cannot tell the difference.

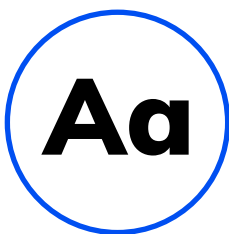


Use

A linear, logical design



Clear color contrast and readable font sizes



A combination of color, shape, and text

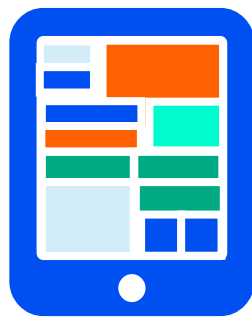


Short texts

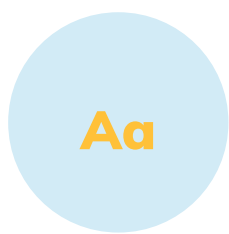


Avoid

Complicated design and menus



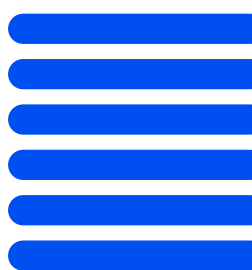
Low color contrast and small font sizes

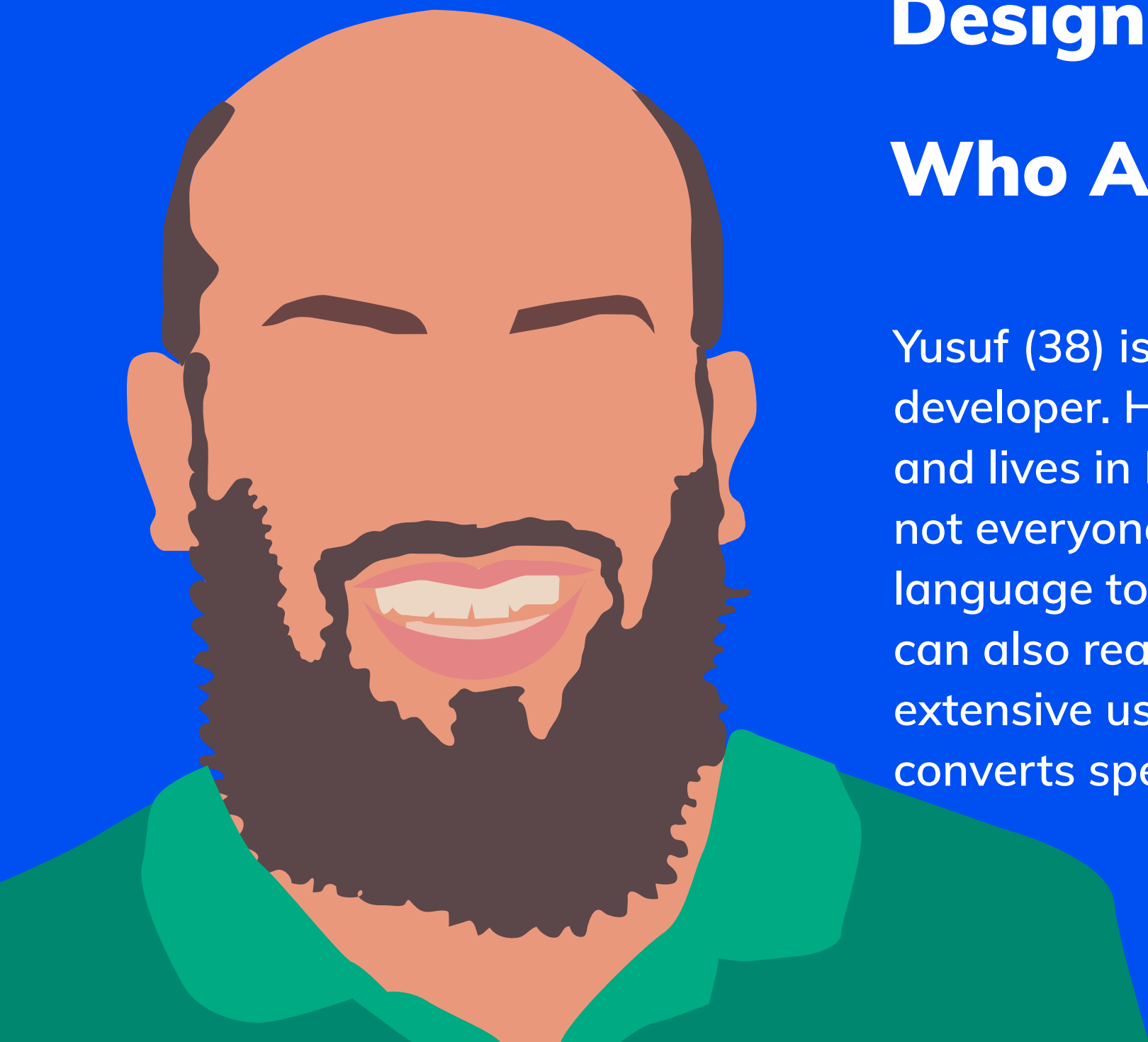


Using only color to convey information



Large blocks of text





Designing for People Who Are Deaf

Yusuf (38) is a programmer at an application and game developer. He became deaf at the age of 12, is married, and lives in Eindhoven. Speaking is difficult for him, and not everyone can understand him well. He uses sign language to communicate with his wife and family. He can also read lips. When talking to others, he makes extensive use of the Live Transcription app. This app converts speech into text.



Use

Simple English

Do this.

Subtitles for videos



Varied content with subheadings, images, and videos



Short texts

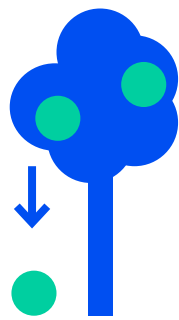


The option to request an interpreter



Avoid

Figurative language and idioms



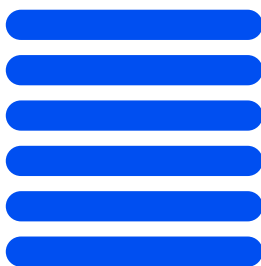
Content available only in audio or video



Complex design and menus



Large blocks of text



Phone calls as the only contact option





Designing for People with physical disabilities

Philipa (22) has a progressive hereditary muscle disease that makes it increasingly difficult for her to control her muscles. She is in pain every day and relies heavily on her medication and assistive devices. She has been declared fully unfit for work, lives with her partner, and has a child. She can walk on her own but is almost always in a wheelchair. When going out, she carefully prepares in advance. What is or isn't accessible at a location? She looks up this information online, but it requires a lot of effort. Excessive scrolling and clicking take too much strength for her hands and arms.

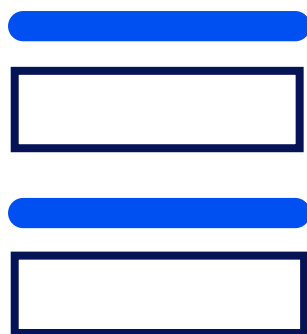


Use

Large, clickable actions



Space around input fields



Design that works with keyboard or voice navigation



Mobile- and touch-screen-friendly designs

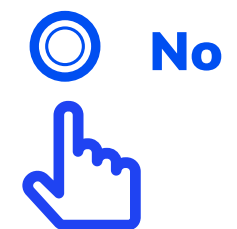


Shortcuts

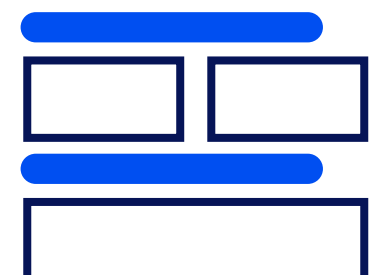


Avoid

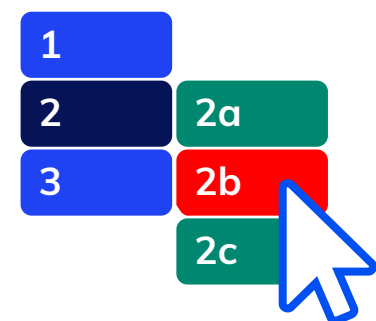
Precise actions



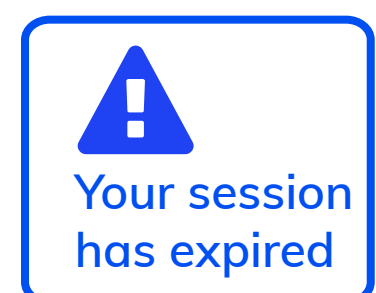
Input fields placed too close together



Dynamic content that requires extensive mouse movement



Short sessions and timeouts



Excessive typing and scrolling





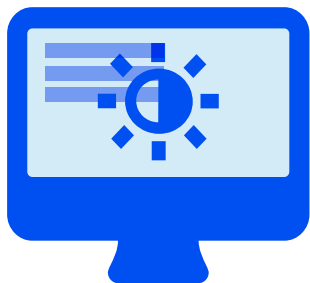
Designing for People with dyslexia

Amina (33) works as a residential counselor in disability care. She exercises a lot and enjoys going to the city with her friends on the weekends. During her studies, dyslexia caused delays and frustration. At work, she has to read and write a lot of handover reports as well as read protocols and other relevant information. It takes a lot of her energy all the time.



Use

The option for users to change color contrast



Left-aligned text and a consistent layout



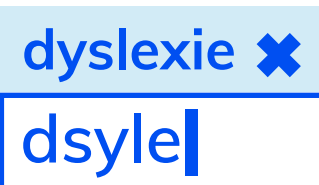
Other formats such as audio and video where possible



Short, clear, and simple content

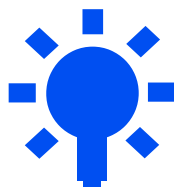


Spelling suggestions and autocorrect



Avoid

The use of flashing animations



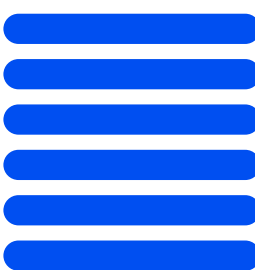
Underlining, italic text, or entire text in capital letters

DON'T
DO THIS.

The need to remember information from previous pages



Large blocks of text



Requiring correct spelling

✗ invalid input

Designing for People with Autism

Wiebe (25) has autism and is studying ICT at Windesheim in Zwolle. He is a third-year student and has been living on his own for two years. Together with a friend, he runs his own business, building and maintaining websites for small entrepreneurs. Wiebe frequently uses his MacBook for both school and his business.

Use

Calm colors



Simple English

Do this.

Short sentences and bullet points



Descriptive buttons

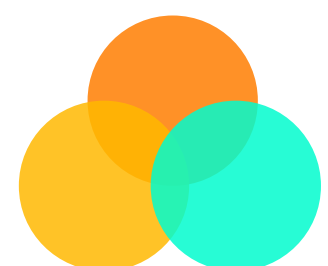


Consistent design

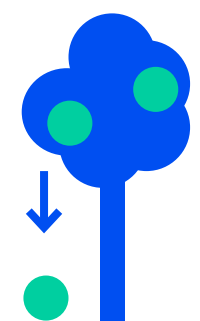


Avoid

Bright colors with low contrast



Figurative language and idioms



Too much information on one page



Unclear or unpredictable buttons



Complicated and cluttered design





Designing for People with Low Literacy

Henk (44) lives in an apartment in Assen with his two children, his girlfriend and three dogs. He works as a mail carrier and could use some financial support. However, he does not apply for subsidies from the municipality or government because he does not understand the texts in information brochures and letters. Filling out forms is extremely difficult for him, and he is afraid of making mistakes.



Use

Short, clear, and simple content



Descriptive buttons



Clear icons



Plenty of white space between lines and paragraphs



Align text to the left



Avoid

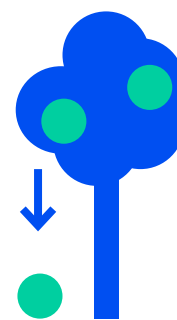
Too much information on one page



Difficult words and long sentences



Figurative language and idioms



Graphs and tables with lots of text



Complicated and cluttered design





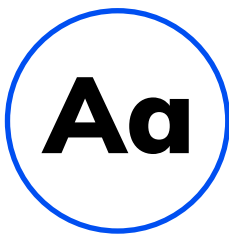
Designing for Seniors

Margriet (67) is now retired. She used to be a lecturer at a university of applied sciences. She lives alone in an assisted living apartment in De Bilt. She has three children and six grandchildren who do not live nearby. She notices that her concentration, memory, and vision are declining, and her hand coordination is also decreasing—she experiences tremors. Surfing online is fun until it becomes too complicated. If she has to order something that requires her to fill out many fields, she gives up. It is too complex and makes her feel insecure.



Use

Good color contrast and readable font size



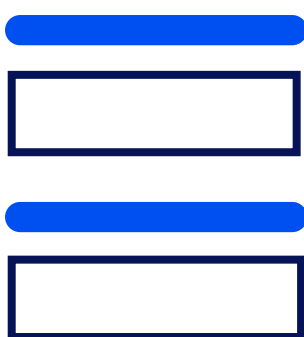
Short, clear, and simple content



Large, clickable actions



Space around input fields

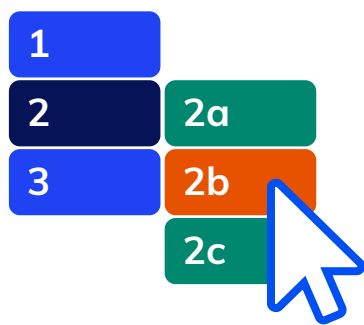


A linear, logical design

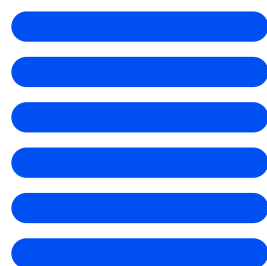


Avoid

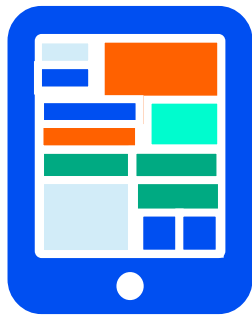
Dynamic content that requires a lot of mouse movement



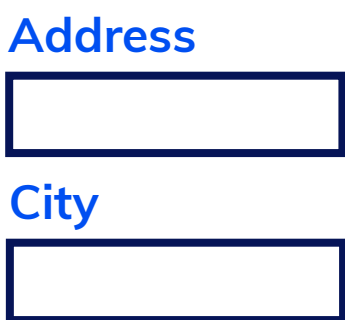
Large blocks of text



Complicated designs and menus



Lots of typing and scrolling



Links and headings without clear information

